BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Malware and Social Engineering

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why are computing standards useful?

2. Where are cloud files stored?
   A) Only in CPU registers
   B) On remote servers
   C) Only in local RAM
   D) Only on optical discs

3. Convert denary 150 to two-digit hexadecimal.

4. Convert hexadecimal 96 to denary. Show your calculation.

Today's objectives

Define malware and identify common types. Explain phishing and other social engineering threats.

Key vocabulary - Read, cover, recall

Malware
Malicious software that harms or compromises systems.

Trojan
Malware disguised as trustworthy software.

Worm
Malware able to replicate and spread to other systems.

Ransomware
Malware that can lock or encrypt data and demand payment.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Malware harms systems

Malware is software designed to harm, disrupt or gain unauthorised access to systems. Different malware types have different behaviours.

Trojan, worm and ransomware

A trojan disguises itself as legitimate software; worms replicate to spread; ransomware may encrypt files and demand payment.

Social engineering

Attackers deceive people into sharing information or taking an unsafe action. A phishing message impersonates a trusted organisation.

Look for warning signs

Unexpected links, urgency and requests for account details are warning signs. Know the threat and explain why sharing credentials is dangerous.

I Do - Worked model

A pupil receives a message pretending to be school IT and asking for login details via a link. Explain the threat and TWO possible consequences. [4 marks]

- The sender pretends to be a trusted school service.
- This is phishing/social engineering [1]: the pupil is directed to a fake page or asked for details [1].
- The attacker may capture login credentials [1] and gain unauthorised access to the pupil's account [1].
- Name the threat, describe the deception and link data theft to access.

We Do - Guided practice

1. A pupil downloads an apparently harmless game that secretly installs malicious software. Explain the threat and two possible effects. [4 marks]

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2. A school finds files have been encrypted and an attacker demands money. Explain this threat and its effect. [4 marks]

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3. A fake text claims a student's account will close unless they enter a password. Explain why this is social engineering. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the training message.

```
suspicious = true
if suspicious then
print("Report message")
endif
```

You Do - Independent exam questions

Q2. Define phishing. [2 marks]

Q3. A veterinary clinic finds its appointment files locked and receives a demand for payment. Explain how ransomware could disrupt the clinic. [4 marks]

Q4. Explain malware and social engineering threats, giving three specific consequences. [6 marks]

Exit Ticket - True or False?

A trojan is always harmless because it looks like legitimate software.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Agreed specifications help hardware and software from different producers interoperate.

2. B - On remote servers

3. 96.

4. (a) 9 x 16 + 6 = 150.

I Do - Explain the model

The sender pretends to be a trusted school service.

This is phishing/social engineering [1]: the pupil is directed to a fake page or asked for details [1].

The attacker may capture login credentials [1] and gain unauthorised access to the pupil's account [1].

Name the threat, describe the deception and link data theft to access.

We Do - Model answers

1. It is a trojan [1] disguised as legitimate software [1]. It could steal personal data [1] or damage/delete files [1].

2. Ransomware is malicious software [1] that encrypts/locks files [1]. Users cannot access the files [1] and an attacker demands payment for restoration [1].

3. It impersonates a trusted service [1] and creates urgency [1], manipulating the victim [1] into disclosing account credentials [1].

Code First - Answer

```
suspicious = true
if suspicious then
print("Report message")
endif
```

Expected output: Report message

You Do - Self-mark

Q2: Deceptive messages impersonate a trusted source [1] to trick users into revealing personal data/credentials [1].

Q3: Ransomware is malicious software [1] that encrypts or locks the clinic's appointment files [1]. Staff cannot access the appointment schedule [1], potentially disrupting bookings and patient care [1].

Q4: Malware is malicious software [1] that can steal data or corrupt files [1]. Social engineering deceives people [1] to reveal information [1]. Examples of consequences: unauthorised access [1] and lost/unavailable data [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.